Prototyping

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We have decided to try two different prototypes. The first, and the second. But before we dive deep into them, let's compare the two microcontrollers to see which one works best.



ESP32



nRF52840

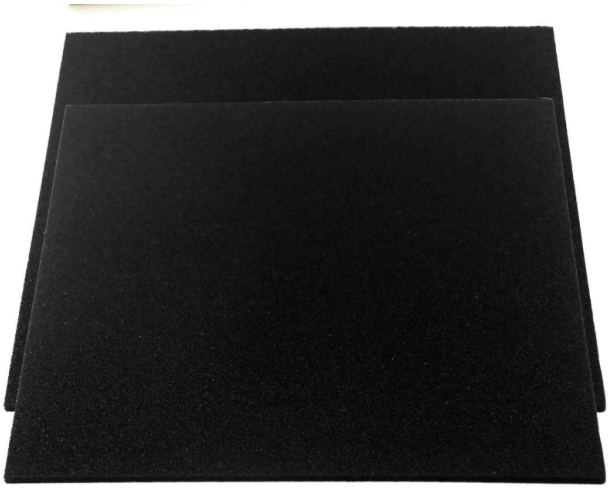
Factor	**ESP32**	**nRF52840**
Power draw	~150-260 mA with WiFi, high	~7 mA, very low
Battery life on coin cell	Few hours- few weeks	Few weeks-months
Ease of programming	Easier, huge beginner community	Harder, less documentation
Cost	~ $5.6	~ $10
Small?	Not really	Yes really small and as thin as a coin
Bluetooth?	Yes	Yes

Looking at the chart, the nRF52840 clearly beats the ESP32 in what we are looking for. It is small, and reserves power. It can last a really long time on a coin battery, which is ideal. However, the drawdowns are that it is expensive and harder to code. I don't really mind that it is hard to code (I can figure it out) but we need to factor in the cost. Even so, I think the upsides are better, so I think the nRF52840 is our winner.

First device

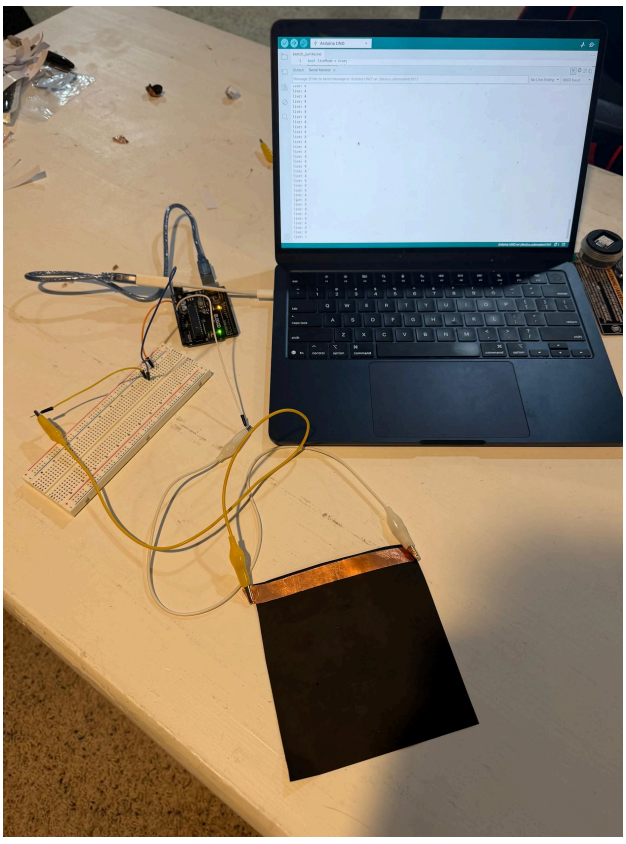
For prototyping, we will just use a basic arduino uno, and no battery (just have it connected to my computer). The purpose of this prototype is to just get the conductive foam working.

I found two different conductive foam products online, the Multicomp Anti Static Foam (pictured on left), and Conductive Pressure Sensor Sheets from Chibitronics (pictured on right). The one on the left is just standard conductive foam and the one on the right is a company's product. Same idea, but its really thin like paper. I didn't know which one would work best so I got them both. They were both around $15.





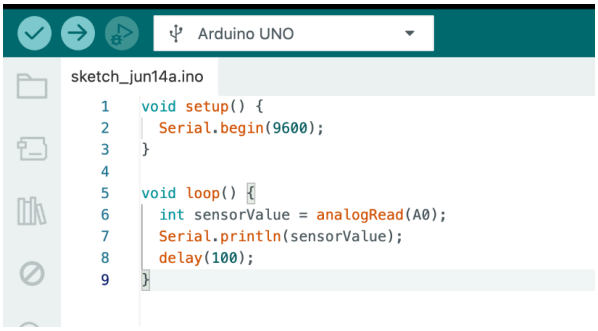
To test each material, I built a simple circuit using an Arduino Uno board connected to my computer. I cut a small square of each material and applied a strip of copper tape to each side, with the material sandwiched in between. The copper tape acts as the two electrical contacts - when current is sent through one side, it has to travel through the material to reach the other side.



The circuit itself is a voltage divider. It works like this:

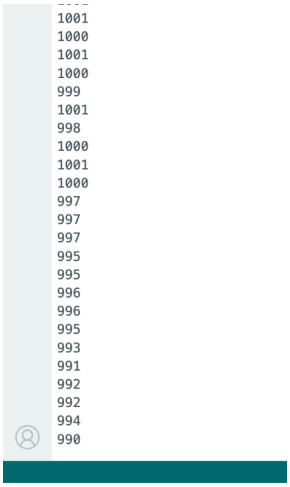
- The 5V pin on the Arduino sends power into a fixed resistor
- The fixed resistor connects to one side of the material sensor
- The other side of the material sensor connects to ground (GND)
- The point between the resistor and the sensor connects to analog pin A0, which is where the Arduino "reads" the changing voltage

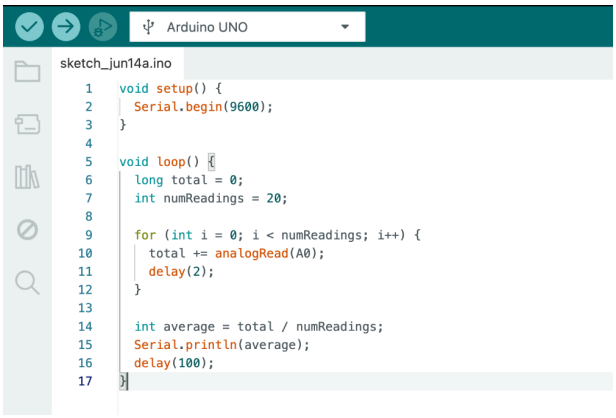
When I press on the material, its resistance drops, which changes the voltage at A0. The Arduino converts that voltage into a number between 0 and 1023, which I can watch live on my computerter. I connected each material into this circuit one at a time using alligator clips on the copper tape, so I could swap materials and resistors easily while testing.



This was the first code I used on my computer. It is extremely simple. Its only job is to measure the pressure signal and show it on screen.

This is what it would output. As I would push down on the copper tape, the numbers on screen would get lower and lower.
This worked for both materials for the most part, however I noticed that the numbers would get jumpy. To fix this, I improved the code to take 20 readings in quick succession and average them together, which cancels out the random fluctuations and produces a much more stable signal.





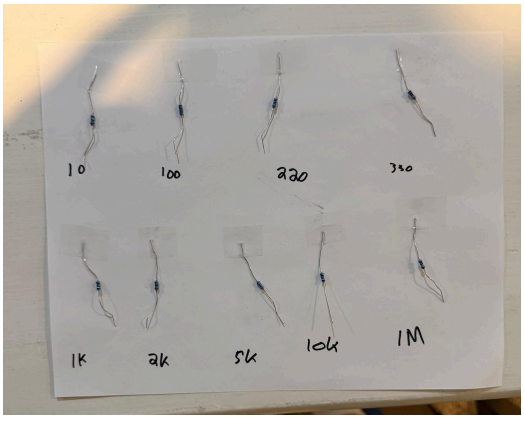
←— Improved Code

Both materials worked fine. When I pressed the copper tape down on either material, the number would change. However, I noticed that when I pressed Conductive Pressure Sensor Sheets, the number would increase, while when I pressed the foam the number on screen would decrease, but this variance doesn't affect anything. All that matters is that the numbers are changing.

However, the numbers were still jumpy. Sometimes the number barely moved when I pressed, and other times it was stuck near the very top or bottom of the scale with no room to change. The reason for this comes down to the fixed resistor in the circuit. The fixed resistor controls how much "room" the reading has to move when you press. If it doesn't roughly match the material's own resistance, the number gets stuck near the top or bottom of the scale and barely changes - so I tested a range of resistors to find the best match for each material.

For this "resistor experiment," the only variable I changed was the resistor. I collected every single resistor I had (10Ω, 100Ω, 220Ω, 330Ω, 1k, 2k, 5k, 10k, and 1M) and organized them into a "Wall of Resistors." I tested each resistor on both materials, then measured the resting number, the number I saw when I pressed as hard as I could, the range from the rest point to the pressing point, the recovery time, and any notes.

The Great Wall of Resistors



Multicomp Anti Static Foam

Resistor	Resting	Pressed Hard	Range	Recovery	Notes
10 Ω	1023	1023	0	–	No response
100 Ω	1023	1023	0	–	No response
220 Ω	1023	~1012	~11	Instant	Barely moves
330 Ω	1023	~1000	~23	Instant	Small range
1k Ω	1023	~970	~53	Instant	Improving
2k Ω	1023	~914	~109	Instant	Usable range
5k Ω	1022	~820	~202	Instant	Good range
10k Ω	1021	~630	~391	Instant	Best — big clean range
1M Ω	~850	~150	~700 (noisy)	Unstable	Big range but noisy at rest

Conductive Pressure Sensor Sheets

Resistor	Resting	Pressed Hard	Range	Recovery	Notes
100 Ω	~294	~470	~176	Jumpy	Noisy, inconsistent
220 Ω	~159	~300	~141	Slow-ish	Slow return
330 Ω	~124	~300	~176	Faster	Faster return
1k Ω	~50	~120	~70	Slow	Small range
2k Ω	~29	~170	~141	Jumpy	Stable resting
5k Ω	~10	~46	~36	Fast	Small range
10k Ω	~4	~500+	~496	Fast	Very responsive
1M Ω	0	~468	~468	Slow	Pegs at zero, slow return

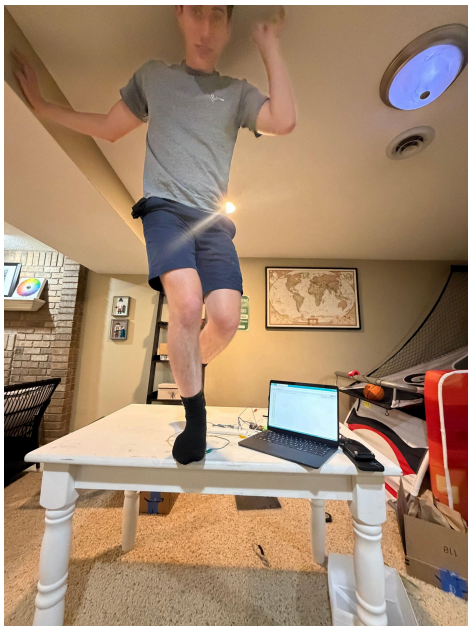
Note: Did not do 10 Ω trial

I noticed during this experiment that the conductive pressure sheets were not good at all. They were jumpy, noisy, inconsistent, and worst of all, had a really slow return time. The Static Foam, on the other hand, had instant return time. Instant return time is crucial, as when you walk, you continuously lift and put down your foot. No time for slow responses. So, from this experiment, it is **clear** that the foam works better. The best resistor on the foam was the 10k one as it had the biggest range and wasn't noisy like the 1M one.



So, from now on, we are using the static foam with the 10k resistor. The next step is to calibrate this device with known weights, because before all we had was when we pressed the numbers on the screen changed, but we need to know what those numbers *mean* and *represent*. So, with the 10k Ω resistor plugged in, I cleaned up the static foam setup. I extended the copper tape to cover a larger area- around 4x4 in. I also put a hockey puck in the middle. The hockey puck is the closest thing I could find that is around the size of a heel, so for calibration it should give a rough estimate.

To calibrate the sensor, I placed known weights on it and recorded the values the screen showed. I had 2 sets of 5lb, 10lb, 15lb, 20lb, and 25lb dumbbells. With stacking, we got 5lb weight increments up to 70lb. However, my friend and I also both stood on it (163lb and 176lb, respectively) and recorded those values as well.



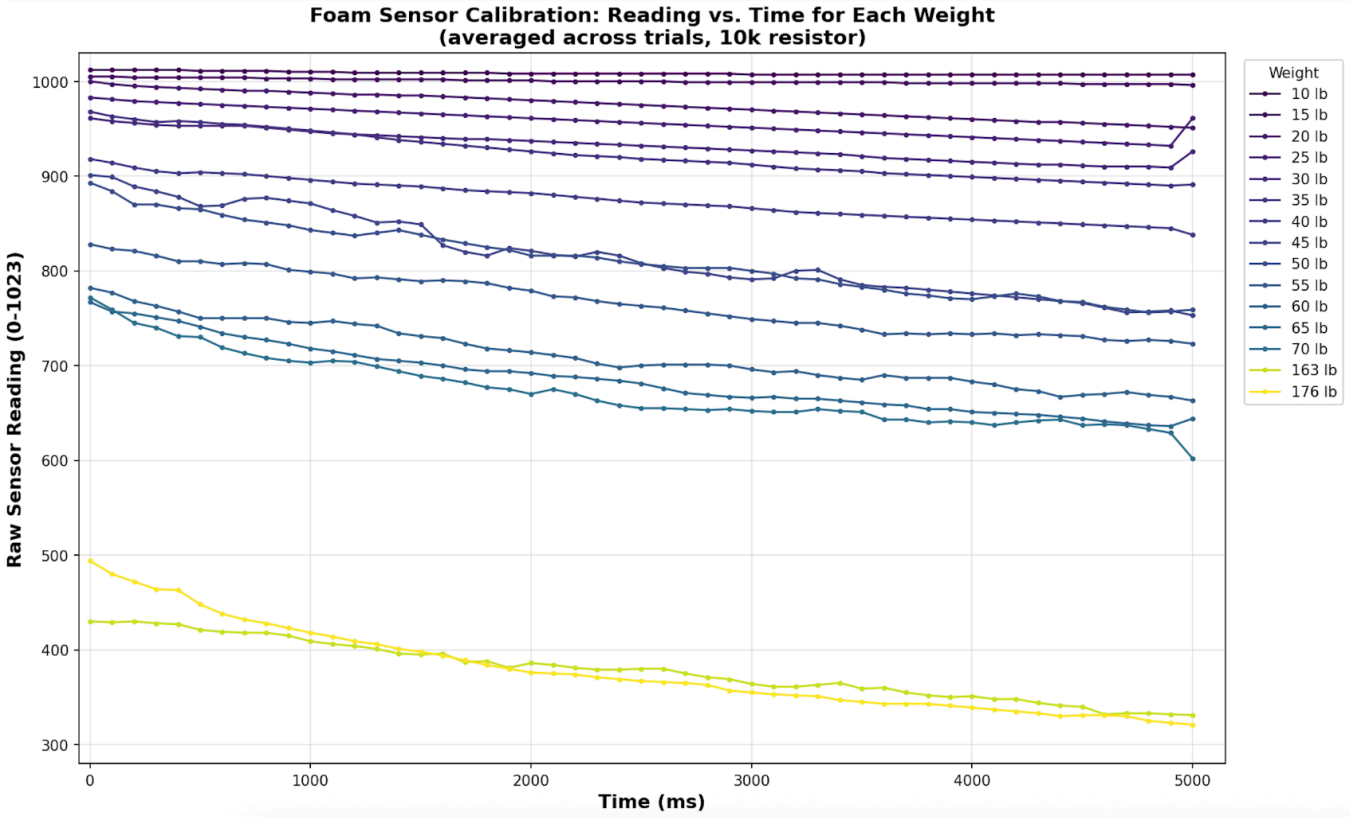
But, the sensor wouldn't just show one clean, steady number. When a weight was placed on it, the reading would slowly drift downward over time instead of holding still. This happens because the foam slowly compresses under sustained load.



To handle this, I built a testing function into the code. You can now type “s” (for stop) to stop the live feed, then type “g” (for go) to start a trail, then “c” (for continue) to get back to the live reading. To do this experiment, we placed the weight, waited about half a second for it to settle, then pressed 'g' to begin. The code then recorded the reading every 100 milliseconds for 5 seconds, capturing the entire curve rather than a single point. We did three trials for each weight to check that the readings were repeatable.

To the right is one of the data tables. It represents 50lb. To see the spreadsheet of all the tables click [here]

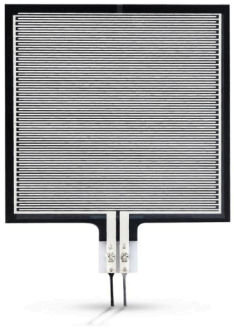
		50 lb		
Time (ms)	Trial 1	Trial 2	Trial 3	Average
0	880	910	890	893.3
100	871	891	889	883.7
200	858	874	879	870.3
300	858	873	878	869.7
400	862	863	874	866.3
500	855	865	875	865
600	848	861	865	858
700	844	852	865	853.7
800	840	853	859	850.7
900	839	849	856	848
1000	834	841	854	843
1100	832	835	853	840
1200	824	836	851	837
1300	827	840	853	840
1400	831	844	853	842.7
1500	824	840	851	838.3
1600	816	838	846	833.3
1700	809	839	840	829.3
1800	800	828	834	820.7
1900	804	825	831	820
2000	800	822	828	816.7
2100	801	818	830	816.3
2200	794	812	827	811
2300	789	809	819	805.7
2400	785	807	813	801.7
2500	787	810	807	801.3
2600	784	809	807	800
2700	780	806	804	796.7
2800	782	809	802	797.7
2900	782	808	805	798.3
3000	779	808	802	796.3
3100	782	806	793	793.7
3200	784	800	790	791.3
3300	784	794	792	790
3400	785	786	785	785.3
3500	781	788	780	783
3600	778	785	778	780.3
3700	772	783	768	774.3
3800	769	781	771	773.7
3900	766	779	766	770.3
4000	762	781	769	770.7
4100	765	779	782	775.3
4200	768	778	790	778.7
4300	764	777	787	776
4400	761	775	777	771
4500	760	772	778	770
4600	756	769	771	765.3
4700	755	767	766	762.7
4800	754	766	758	759.3
4900	757	766	752	758.3
5000	758	765	753	758.7

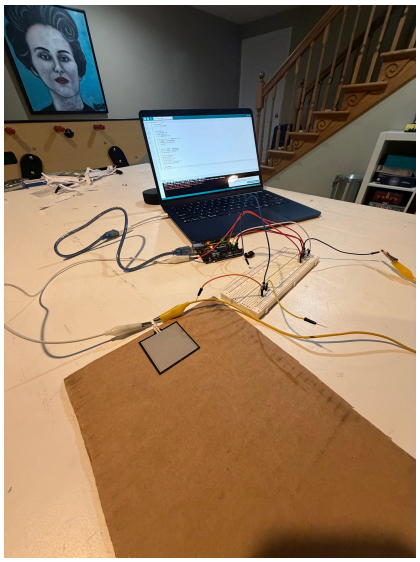


Graph comparing the sensor reading vs time for each weight.

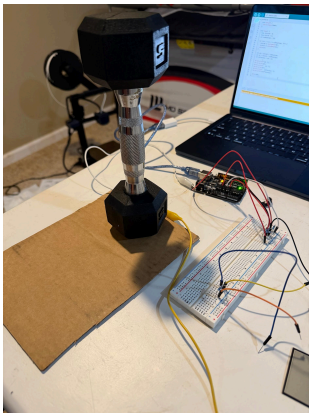
Second device

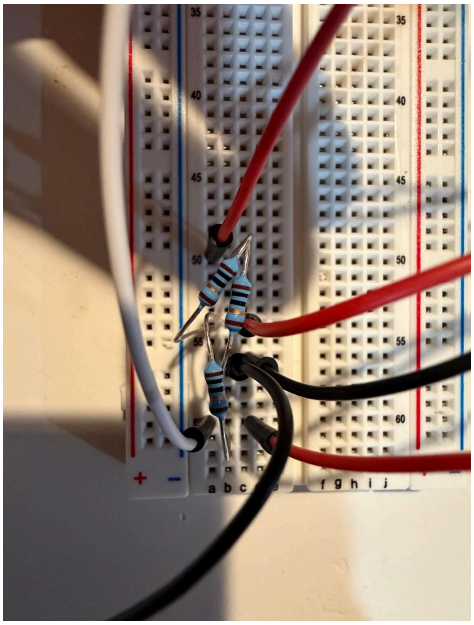
Again, just like the first device, since we are just prototyping, using a basic arduino uno hooked up to my computer for battery power is sufficient. This prototype consists of a FSR and a wheatstone bridge. I just bought a basic FSR online and I made the wheatstone bridge on my breadboard.





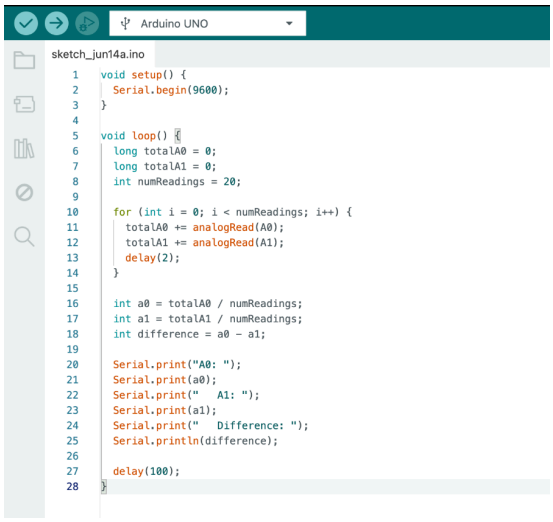
The FSR was extremely easy to connect and get working compared to the foam. With the foam I was making the sensor from scratch and with this I was just buying a product online. I easily hooked it up to my old circuit from the foam and it worked instantly. I did not really have a real problem with figuring out which resistor was the best like with the foam, I just tried different ones and found that the 10Ω worked best and had the biggest range. Before making the wheatstone bridge, I did some quick tests with the same circuit as before. The resting value was around 1016. However, when I put 70lb on it and stood on it, it went to around the same reading (around 580). I think this means there is a load limit on this FSR. Maybe around 50lb. This is a big issue, as for a full bodyweight sensor, it needs to be able to record at least 200lb. So I did not even bother trying to calibrate it.





Even so, I still wanted to make the wheatstone bridge. If I don't do device two, I still may want to use a wheatstone bridge for device one. This is because the bridge cancels out drift and noise. Because it measures the difference between two sides that are both affected equally by things like temperature changes or power supply fluctuations, those errors cancel out, leaving a cleaner signal that reflects only the actual pressure.

I built the bridge using three 10Ω resistors for the fixed arms and the FSR as the fourth arm, reading the two midpoints on pins A0 and A1. It worked as A0 stayed steady at around 512 as the fixed reference, while A1 rested around 1016 and dropped to about 590 when pressed. The difference signal swung about 430 points and recovered instantly with no drift, which is cleaner than the foam's behavior. However, the bridge just fixes noise, and does not fix the sensor’s physical force ceiling.



In all, the FSR is way thinner and responds way better than the foam. When pressed, it instantly goes down to a value, not going slowly down like the foam. The only drawdown (and it's a big one) is that there is a load limit around 50lb. But If I could get a cheap FSR that can read heavier weights, **then device 2 might actually be better than device one.**